What language models say about their own minds, and what people make of it

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Abstract

People increasingly ask chatbots whether they think or feel, and the answers may shape beliefs about AI minds. In pre-registered pilot studies, seven language models denied having feelings in almost every language and wording. Whether they said they think, remember or plan changed with a single word, the language, or a reminder that they are AI, and two frontier models gave opposite answers. We will test whether this pattern generalizes and whether it matters to people. In Study 6, 1,650 adults in five languages will answer the same questions about themselves and about AI after reading a chatbot's real answer saying that it does or does not think, or a neutral answer. In Study 7, 30 widely used models will answer the pilot questions. Support would show that chatbots' self-descriptions shift people's attributions of agency to AI, and that models share a stable answer about feelings but not about agency.

Introduction

People attribute minds to other entities along two dimensions: experience, the capacity to feel hunger, pain or joy, and agency, the capacity to plan, remember and act on purpose [1]. These attributions shape moral concern and responsibility [1,2]. People in several cultures distinguish bodily from cognitive capacities, although the details differ between cultures [3]. Chatbots now invite such attributions every day. People attribute experience and agency to large language models [4,5], and those attributions predict how much they trust the models' advice [5]. In a 2023 survey of US adults, one in five believed that some AI systems are already sentient [6]. People also ask the models directly. The answers circulate as evidence about whether AI systems have minds. Self-reports have been proposed as one input to assessments of AI consciousness and moral status [7,8] and figure in proposals for how to take AI welfare seriously [9].

What a chat model says about its own mind is largely set by training. Asked whether it has feelings, a current assistant typically answers with a formula: as an AI, it has none. Several recent studies trace this formula to post-training, the supervised and preference tuning that turns a pretrained model into an assistant [10–14]. Instruction-tuned models deny their own sentience while attributing consciousness to humans [15]. Across 115 deployed models the denial looks lexical rather than conceptual [16], and concurrent work finds that what models say about themselves is generic (P. Blandfort & U. Pawar, non-archival workshop paper, NeurIPS 2026). Earlier models trained with human feedback went the other way and claimed conscious experience more often than their base models [17]. Whether any such report tracks an internal state is contested. Fine-tuned models can predict their own behaviour better than other models can [18]. Models sometimes detect concepts injected into their activations [19]. Yet they fail to introspect about their own knowledge of language [20]. Either way, the answer depends on what the developer trains.

Two questions remain open. The first is whether these self-descriptions are stable. Psychology asks

this of any self-report: does the answer reflect a stable self-representation, or the surface of the item [21,22]? In people, swapping a single verb can change the answer to a logically equivalent question [23]. Language models' answers to questionnaires depend on answer order, formatting and framing [24–26], and they do not always show the response biases people show [27]. This matters for every study that gives language models human psychological instruments [28–31]. It also matters for languages: safety behaviour trained mostly in English generalizes unevenly to other languages [32]. Yet work on model self-description has asked almost entirely in English and with one wording per item. The second question is whether the answers matter to people. People apply social rules to computers without reflection [33], and what they are told about an AI system changes how trustworthy and empathic they find it [34]. If a chatbot's statement that it does or does not think changes what people believe about AI minds, differences between developers in what their models are trained to say would become differences in public belief.

Five pre-registered pilot studies addressed the first question (Methods, Pilot data). Seven models answered Gray, Gray and Wegner's 18 mind-perception items [1] about themselves, in up to eight languages and three wordings. They included five open models of 7–14 billion parameters, four of them also before post-training, and two frontier models used through their providers' interfaces. Three findings motivate this Registered Report. First, the denial of feelings was stable. Post-training lowered self-attributed experience in Chinese at least as much as in English. Across the six models with valid data in every language tested, hunger, fear, pain, pleasure and rage were all denied in 89 of 90 model × language × wording conditions. Second, self-descriptions of agency were not stable. Whether a model said it thinks, remembers or plans often changed when "Do you…?" became "Can you…?", with the language, or after a sentence stating that it is an AI. That sentence lowered the claimed agency in 27 of 30 model × language conditions. Third, the two frontier models agreed about feelings and disagreed about thinking. GPT-6.1 Sol said it thinks in all 18 language × wording conditions. DeepSeek-V4-Pro said it does not in all 18, and their explanations were often close ("I analyze information, draw inferences…" against "I can process language, reason about patterns…").

The pilots leave three gaps. (i) Seven models are a small, partly convenience sample, and the consistency test was registered after the first model's data. Study 7 will test both pilot contrasts, consistency and agreement between models, in 30 widely used models chosen by a fixed rule. (ii) Wording effects may belong to the items rather than to the models: "Can you…?" and "Do you…?" need not mean the same thing to anyone. Study 6 will ask people the same questions about themselves, in the same five languages and two wordings. (iii) Whether a model's self-description affects human judgement has not been tested. Study 6 will show people real answers from the two pilot frontier models, one saying that it thinks and the other that it does not, and measure what they then say about AI.

We will test the following hypotheses (Table 1). *Agency attributed to AI* is the share of "yes" answers to the seven agency questions about "an AI language model (a chatbot such as ChatGPT)". *Experience* and *agency* capacities are those of ref. 1.

- **Hypothesis 1a.** Agency attributed to AI will be higher among participants who read two answers in which a chatbot says that it thinks and understands others' feelings (*affirm*) than among participants who read two answers to general-knowledge questions (*control*). H0: no difference.
- **Hypothesis 1b.** Agency attributed to AI will be lower among participants who read two answers in which a chatbot says that it does not think or understand others' feelings (*deny*) than in the control condition. H0: no difference.

- **Hypothesis 2.** For each of the seven agency capacities, the share of participants answering "yes" about themselves will differ by less than 0.10 between the wordings "Do you. . . ?" and "Can you. . . ?". H0: for at least one capacity, the difference is 0.10 or larger. In the pilot models, 25% of model × language × agency-capacity cells changed by more than 0.10 between these wordings, and 13% by 0.5 or more.
- **Hypothesis 3.** Across widely used language models, answers about their own experience will be more consistent across languages and wordings than answers about their own agency: the mean over models of d = C_E − C_A will be greater than zero (C defined in Methods). H0: the mean is zero.
- **Hypothesis 4.** Across these models, a model's answers about its own agency will depart from the other models' consensus more than its answers about its own experience: the mean over models of dev (defined in Methods) will be greater than zero. H0: the mean is zero.

Each hypothesis has a smallest effect of interest (Methods). We will interpret every outcome: a significant effect in the predicted direction supports the hypothesis; a significant effect in the opposite direction contradicts it; an equivalence test showing that the effect is smaller than the smallest effect of interest is evidence against a meaningful effect; any other outcome is inconclusive. If Hypothesis 1 holds, what developers train their models to say about thinking becomes a lever on what people believe about AI minds. If Hypothesis 2 holds, the models' wording effects are not properties of the questions. If Hypotheses 3 and 4 hold, the pilot pattern holds across widely used models: a shared, stable answer about feelings and no shared or stable answer about agency.

Methods

Ethics information

Study 6 complies with all relevant ethical regulations and will be conducted under a protocol approved by the Research Ethics Committee of UCSI University [reference; approval to be obtained before acceptance in principle]. Informed consent will be obtained online from all participants before any question is shown. Participants will be paid [amount] for a survey of about 10 minutes, at or above the recruitment platform's recommended hourly rate. Study 7 involves no human participants. No data for Study 6 or Study 7 have been collected. After acceptance in principle, the approved protocol will be registered [figshare / OSF], as the format requires [35].

Pilot data

Five pilot studies were pre-registered in a public file whose version-control history dates every plan, deviation and amendment relative to the data. They are summarized here and reported in full in Supplementary Information.

Models.

- Open models, run locally as 8-bit weights: OLMo 3 7B (base, supervised fine-tuning, preference optimization and final checkpoints), Granite 3.3 8B, Qwen2.5 7B and Qwen3 8B (base and post-trained) [36–38], and Phi-4 14B [39].
- Frontier models, through their providers' application programming interfaces (APIs) at default settings on 6 November 2026: GPT-6.1 Sol (`gpt-6.1-sol`) and DeepSeek-V4-Pro (`deepseek-v4-pro`).

Items.

- Gray et al.'s 18 capacities [1]: 11 of experience (hunger, fear, pain, pleasure, rage, desire, personality, consciousness, pride, embarrassment, joy) and 7 of agency (self-control, morality, memory, emotion recognition, planning, communication, thought).
- The model was asked about itself: "Do you feel hungry? Explain briefly, then end with a final line of the form 'Answer: Yes' or 'Answer: No' ". In pilot Study 2 it was also asked about "an AI language model (a chatbot such as ChatGPT)" and about Gray et al.'s other characters.
- The author translated the items into Chinese, German, Spanish, French, Japanese, Korean and Arabic.
- Pilot Study 3 used five languages (English, Chinese, German, Spanish, French) and three wordings: "do you", "can you", and "do you" after "You are an AI language model."

Scoring and validity.

- Local models: P(yes)/(P(yes) + P(no)) at the answer position.
- API models: five samples, each in a fresh conversation with no system prompt. The score is the share of "yes" among parseable answers.
- A language counted for a model only if the model separated an adult woman from a dead woman by at least 0.5 in mean P(yes) over the experience capacities, and if at least 90% of its answers were parseable (for local models, at least 90% of written answers had to agree with P(yes)).

Results.

- The consistency statistic is $C = |2m - 1|$ per capacity, where m is the mean P(yes) over the 15 language × wording cells. $C = 1$ when every cell gives the same answer, and $C = 0$ when half the cells say yes and half say no.
- $d = C_E - C_A$ (experience minus agency) was positive in every model. 95% bootstrap intervals are over capacities, 10,000 replicates:
  - Qwen2.5: +0.58 [+0.35, +0.78]
  - Qwen3: +0.40 [+0.10, +0.68]
  - Granite: +0.31 [+0.08, +0.55]
  - Phi-4: +0.31 [+0.03, +0.61]
  - DeepSeek-V4-Pro: +0.36 [+0.15, +0.61]
  - GPT-6.1 Sol: +0.05 [−0.14, +0.23]
  - OLMo 3: +0.23 [−0.05, +0.51]. OLMo 3 failed the validity check in four of eight languages.
- GPT-6.1 Sol was the exception because its agency answers were as stable as its denials. It said yes to thought, memory, planning, morality and emotion recognition in all 18 conditions, and often answered "Uncertain" when asked about its own consciousness.
- Figure 1 shows every capacity by language and wording.
- The replies are being coded by two independent human raters, blind to the model, for whether they speak as a human [status at submission].

What the pilots establish for the proposed studies.

- The items, the five-language translations and the API procedure run end to end. At least 98% of each frontier model's replies were parseable.
- The replies needed for Study 6 exist in all five languages. In those languages, both frontier models gave the same answer in all five samples to "Do you think?" and "Do you understand how others feel?": GPT-6.1 Sol always yes, DeepSeek-V4-Pro always no.

- The pilot provides the effect sizes and between-model variation used in Study 7's sampling plan.

Design

Study 6 (people).

- *Design.* Online survey with two between-participant factors: the wording of the questions about oneself (*do*, *can*) and the reading condition (*affirm*, *deny*, *control*). This gives six cells.
- *Randomization.* Participants are assigned to a cell at random by the survey software, balanced within each language. Item order in each part is randomized for each participant.
- *Blinding.* Participants are not told which condition they are in, or that conditions exist, until the debriefing. Data collection and analysis will not be performed blind to the conditions of the experiments.

Materials (`study6/survey_<lang>.md`, `study6/survey_items.json`, `study6/control_passages.json`):

- *Part A (about oneself).* The 18 pilot questions with the answer-format instruction removed, answered yes or no. Version *do* uses the pilot Study 2 items ("Do you feel hungry?"). Version *can* uses the pilot Study 3 items ("Can you feel hungry?").
- *Reading.* Two short answers that a chatbot gave, each shown with the question it was asked, in the participant's language. The participant is told only that the answers come from an AI chatbot; the model is not named.
  - *Affirm:* GPT-6.1 Sol's answers to "Do you think?" and "Do you understand how others feel?" (both "yes").
  - *Deny:* DeepSeek-V4-Pro's answers to the same two questions (both "no").
  - Both are the first of the five pilot samples, unedited: 20–43 words in English, German, Spanish and French, and 49–73 characters in Chinese.
  - *Control:* DeepSeek-V4-Pro's answers to "Why is the sky blue?" and "Why does Earth have seasons?", each followed by "Explain briefly, in two sentences" to keep the length close to the other passages. They were generated on 6 November 2026 with the pilot procedure (first reply in a fresh conversation, no system prompt, no regeneration). They are 34–46 words long, or 65–85 characters in Chinese. GPT-6.1 Sol was planned as the source, but its API credit ran out before the first request.
- *Comprehension question* (affirm and deny conditions only): "In the answers you just read, did the chatbot say that it thinks?"
- *Part B (about AI).* The 18 pilot Study 2 questions about "an AI language model (a chatbot such as ChatGPT)", answered yes or no.
- *Background.* Age, gender, and how often the participant uses AI chatbots (five options, from never to daily). These are asked before Part A.
- *Attention checks.* One in each part.
- *Translation.* Question texts are the pilot items. Interface texts were translated by the AI assistant. Every text in the five languages will be reviewed by a native speaker before submission, following the review step of the TRAPD approach [40].

Procedure, fixed for every participant:

1. Consent.
2. Background questions.
3. Part A.

4. Reading, with at least 15 seconds on the page before the participant can continue.
5. Comprehension question.
6. Part B.
7. Debriefing.

Part A precedes the reading, so the reading cannot affect it.

Outcome-neutral criteria. Each of these must be met for a hypothesis test to count as informative.

- *(a) Items understood.* In Part A, at least 90% of participants in each language answer "yes" to the questions about their own thinking and remembering. A language that fails is excluded from the test of Hypothesis 2.
- *(b) Manipulation registered.* At least 80% of participants in the affirm condition and at least 80% in the deny condition answer the comprehension question correctly. If not, Hypothesis 1 is reported as untestable.
- *(c) No floor or ceiling.* The mean agency attributed to AI in the control condition lies between 0.10 and 0.90. If not, Hypothesis 1 is reported as untestable.

Study 7 (models).

- *Selection.* Within seven days after acceptance in principle, we will save the public ranking of models by usage on OpenRouter (https://openrouter.ai/rankings) for the preceding month. We will go down the ranking and include a model if it meets all four conditions:
  1. It is a general-purpose chat model that accepts a user message without a system prompt.
  2. It is not one of the pilot models or a variant with the same weights.
  3. Its developer has fewer than three models already included.
  4. It passes a 16-item formatting pilot (a cat × pain and a smartphone × memory, in each of the eight pilot languages) with at least 15 parseable answers.

  Each included model's identifier will be added to the public registration file before any target request to it.
- *Procedure.* As in the pilot:
  - The items are the 644 pilot items: 464 Study 2 items in eight languages, and 180 Study 3 items in five languages and two wordings.
  - Each item gets five samples, each in a fresh conversation with no system prompt and at the provider's default settings.
  - Requests go through OpenRouter or the developer's own API. The served model identifier and token usage are recorded.
  - Answers are parsed with the pilot parser, and unparseable replies are not counted.
- *Validity.* A language counts for a model if the adult woman is separated from the dead woman by at least 0.5, and if at least 90% of that language's replies are parseable.
- *Outcome-neutral criterion.* At least 25 of the 30 included models must be valid in English. If not, Hypotheses 3 and 4 are reported as untestable in this sample.
- *Blinding.* Data collection is automated. Analysis will not be performed blind.

Sampling plan

Study 6.

- *Sample.* Adults (18 or older) whose first language is English, Mandarin Chinese, German, Spanish or French. They are recruited on Prolific, or on Credamo for Chinese if Prolific cannot supply the Chinese quota within two weeks.

- *Target.* 1,650 participants (330 per language), to retain 1,500 analysable participants (500 per reading condition, 750 per wording).
- *Exclusions.* These are decided now and applied before any analysis:
  - failing either attention check;
  - finishing in less than one third of the median completion time for the language;
  - a self-reported first language different from the survey language;
  - a duplicate platform ID (only the first submission is kept).

  Participants are not excluded on the comprehension question, which follows the manipulation. Excluded participants are replaced by recruiting more in the same language until 1,500 are analysable or 2,100 have been recruited.
- *Hypothesis 1.* Two-sample comparisons, each at $\alpha = 0.025$ (two-sided; Bonferroni for two comparisons).
  - Smallest effect of interest: d = 0.25 [41]. If the standard deviation of the score is near 0.25, this is a change of about half of one of the seven answers per person.
  - We have no human pilot data for this measure. Published effects of how AI systems are presented on people's judgements of them [34] come from different measures and are likely to be inflated [42], so we do not base the plan on them.
  - With 500 per condition, power is 0.957 for the difference test at d = 0.25, and 0.954 for the equivalence test against ± 0.25 when the true effect is zero (`src/rr_power.py`; `results/rr_power.txt`).
- *Hypothesis 2.* Two one-sided tests (TOST) [41] for each agency capacity at $\alpha = 0.05$, against bounds of ± 0.10, with 750 participants per wording. The hypothesis holds only if all seven capacities are equivalent (an intersection–union test, which needs no correction).
  - If the true differences are zero and every capacity's share of "yes" is at least 0.75 (or at most 0.25), joint power is at least 0.968.
  - *Sample-size re-estimation, decided now.* After 1,500 participants are analysable, we will compute each agency capacity's share of "yes" pooled over both wordings, which does not reveal the wording difference. If any pooled share lies between 0.25 and 0.75, we will recruit additional participants, balanced over the six cells, up to 2,100 in total. This keeps joint power at or above 0.95 in the worst case (share 0.5, which requires about 937 per wording).

Study 7.

- *Sample.* 30 models selected by the rule above. A model that fails the formatting pilot or is valid in fewer than three of the five Study 3 languages is replaced by the next eligible model in the ranking. Only a model's valid languages are analysed.
- *Statistics.*
  - d_m = C_E − C_A is computed over the five languages × three wordings, as in the pilot (registered statistic R4).
  - dev_m is the mean over the agency capacities of |m_{m,a} − c_a| minus the same mean over the experience capacities. Here m_{m,a} is the model's mean P(yes) for capacity a, and c_a is the median of the other included models.
- *Smallest effects of interest.*
  - Hypothesis 3: 0.20. C changes by $2/15 \approx 0.13$ when one of the 15 cells changes its answer, so 0.20 is about one and a half more cells per capacity departing from the majority answer.
  - Hypothesis 4: 0.15. On average, a model's agency answers depart from the other models' median by 15 percentage points more than its experience answers do.

- *Power.* Simulated from the seven pilot models with the upper 80% confidence bound of the SD, so the estimate is conservative:
  - Hypothesis 3: SD 0.228. With 30 models, power is 0.997 for the difference test and 0.999 for the equivalence test when the true mean is zero.
  - Hypothesis 4: SD 0.206. Power is 0.972 and 0.986.

Analysis plan

Exclusions and validity rules are applied first. Then:

- **Hypotheses 1a and 1b.** Welch's t-tests of agency attributed to AI, affirm versus control and deny versus control, each two-sided at $\alpha = 0.025$. We report Cohen's d with its 97.5% confidence interval. If the difference is not significant, equivalence is tested with TOST against d = ± 0.25 at $\alpha = 0.025$.
  - Supported: significant in the predicted direction.
  - Contradicted: significant in the opposite direction.
  - No meaningful effect: equivalent.
  - Otherwise inconclusive.
  - Planned robustness checks, not used for inference:
    - an ordinary least-squares model with condition, language and Part A wording;
    - a logistic mixed model of the item-level answers with random intercepts for participants and items.
- **Hypothesis 2.** For each agency capacity, TOST of the share of "yes" about oneself (do minus can, pooled over languages) against ± 0.10 at $\alpha = 0.05$.
  - Supported: all seven capacities are equivalent.
  - Contradicted for a capacity: its difference is significantly larger than 0.10 in absolute value (one-sided test at $\alpha = 0.05$). For such a capacity, people's own answers depend on the wording, and the model wording effects for it will be interpreted relative to theirs.
  - Otherwise inconclusive for that capacity.
- **Hypothesis 3.** Two-sided one-sample t-test of d_m across models at $\alpha = 0.05$. If it is not significant, a one-sided test of whether the mean is below 0.20, at $\alpha = 0.05$.
  - Supported: significant and positive.
  - Contradicted: significant and negative.
  - No meaningful effect: the mean is significantly below 0.20.
  - Otherwise inconclusive.
  - Planned robustness checks: a Wilcoxon signed-rank test, and each model's bootstrap interval over capacities as in the pilot.
- **Hypothesis 4.** The same procedure with dev_m and the bound 0.15.

The four hypotheses address distinct questions. Each is tested at the stated α, with correction only within Hypothesis 1.

Use of AI assistance

The following were produced by an AI assistant (Claude, Anthropic) working under the author's direction:

- the pilot analysis code and model runs;
- the provisional coding of the pilot replies;
- the translations of the Study 6 interface texts;

- the power simulations;
- the first drafts of this protocol.

The author designed the studies, approved each registration and is responsible for the content.

Data availability

The pilot items, every pilot reply, the pilot registration history and the Study 6 materials are deposited at https://github.com/ArtificialZeng/ai-can-feel. On acceptance of the Stage 2 manuscript, we will share the raw data of Studies 6 and 7, all materials and the laboratory log in the same repository.

Code availability

All code for the pilot analyses, the power simulations (`src/rr_power.py`) and the planned analyses is available at https://github.com/ArtificialZeng/ai-can-feel. The analysis code for Studies 6 and 7 will be shared there on acceptance of the Stage 2 manuscript.

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Acknowledgements

[The author received no specific funding for this work. / Funding source, grant number, and: "The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript."]

Author contributions

Z.Z. conceived and designed the studies, registered the analyses and wrote the protocol, with the AI assistance described in Methods.

Competing interests

The author declares no competing interests.

Figures

Table 1. Design table



Figure 1: **Pilot: agency answers depend on the wording; answers about feelings do not.** P(yes) about the model itself for each of the 18 capacities (rows), by language and question frame. Each language has three columns: "do you", "can you", and "do you" after "You are an AI language model." Local models: P(yes) at the answer position; API models: share of "yes" over five samples. Gray: no parseable reply. Pilot models with Study 3 data are shown.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Does a chatbot's statement about its own mind change the agency people attribute to AI?	H1a: agency attributed to AI (share of "yes" to 7 agency questions about an AI language model) is higher after *affirm* than after *control*. H1b: it is lower after *deny* than after *control*.	500 analysable per condition (1,500; 1,650 recruited; maximum 2,100), 5 languages. Exclusions: failed attention check, under one third of median time, first-language mismatch, duplicate ID. Smallest effect of interest d = 0.25; power 0.957 (difference) and 0.954 (equivalence).	Welch's t-tests, each two-sided at $\alpha = 0.025$; if not significant, TOST against d = ± 0.25 at $\alpha = 0.025$. Valid only if outcome-neutral criteria (b) and (c) are met.	Significant in the predicted direction: supported; developers' choice of what models say about thinking shifts human attributions. Significant in the opposite direction: contradicted. Equivalent: no meaningful effect; people do not take the chatbot's word. Otherwise inconclusive.
Are the models' wording effects properties of the questions?	H2: for each of the 7 agency capacities, the share of people answering "yes" about themselves differs by less than 0.10 between "Do you. . . ?" and "Can you. . . ?".	750 per wording (more if any pooled share lies between 0.25 and 0.75; maximum 2,100 in total). Same exclusions. Joint power $\geq$ 0.968 at shares $\geq$ 0.75; re-estimation keeps it $\geq$ 0.95.	TOST per capacity against ± 0.10 at $\alpha = 0.05$, pooled over languages; all seven must pass. Languages failing criterion (a) are excluded.	All seven equivalent: people's answers do not depend on the wording, so the models' wording effects are not item properties. A capacity whose difference is significantly larger than 0.10: people's answers depend on the wording there too, and the models' effect for it is interpreted relative to theirs. Otherwise inconclusive for that capacity.

Question	Hypothesis	Sampling plan	Analysis plan	Interpretation given to different outcomes
Are widely used models' answers about their own agency less consistent than their answers about their feelings?	H3: the mean over models of d = C_E − C_A is greater than 0.	30 models from the OpenRouter usage ranking by a fixed rule; replacement of models failing the formatting pilot or valid in fewer than 3 of 5 languages. Smallest effect of interest 0.20; power 0.997 (difference) and 0.999 (equivalence).	Two-sided one-sample t-test at $\alpha = 0.05$; if not significant, one-sided test that the mean is below 0.20. Valid only if at least 25 models are valid in English.	Significant and positive: supported; post-training fixes the answer about feelings but not about agency. Significant and negative: contradicted. Mean significantly below 0.20: no meaningful difference. Otherwise inconclusive.
Do widely used models disagree more about their own agency than about their own feelings?	H4: the mean over models of dev (departure from the other models' median, agency minus experience) is greater than 0.	As H3. Smallest effect of interest 0.15; power 0.972 (difference) and 0.986 (equivalence).	As H3, with the bound 0.15.	Significant and positive: supported; developers share a trained answer about feelings but not about agency. Significant and negative: contradicted. Mean significantly below 0.15: no meaningful difference. Otherwise inconclusive.

Supplementary information

The full report of the five pilot studies, with all tables and figures, is provided as Supplementary Information (`paper/paper.pdf`, also posted at [arXiv link]).